

AIM: Qualitative detection of special elements in organic compound
(Nitrogen, sulphur and halogens are usually detected by Lassaigne's test)

PHYSICAL CHARACTERIZATION:

- a) State:
- b) Texture:
- c) Color:
- d) Odor:
- e) Melting or Boiling point:

SODIUM EXTRACT PREPARATION:

Take a small pea size freshly cut and dried Na metal and place in a fusion tube. Heat the fusion tube (holding with a tong) gently, so that Na melts. Add about 0.2 gm of the compound to be tested in it and heat the mixture to redness. Plunge the fusion tube immediately into about 8-10 ml of distilled water kept in small clean mortar. Grind the whole mass thoroughly with the help of pestle and filter it. The filtrate is called sodium extract, which is being used to detect nitrogen, sulphur & halogens.

TEST FOR SPECIAL ELEMENTS (NITROGEN, SULPHUR & HALOGENS):

Experiment	Observation	Inference
1. Take about 5ml of Na extract in a test tube. Add 5ml of freshly prepared dilute ferrous sulphate solution boil it for a few minute, Add two – three drops of dil. FeCl_3 solution and then excess sulfuric acid into the mixture.	A blue or green coloration or prussian blue precipitate. OR If red coloration is produced.	Nitrogen is present. N & S both Present.
2 Take a few ml. of Na-extract in a test tube. Add a few ml of freshly prepared very dilute solution of Na nitroprusside to it.	Brilliant purple colours appear and slowly vanish on standing.	Sulphur is present.

3. Beilstein test for Halogen: Heat a piece of cu- foil in non- luminous Bunsen flames till it ceases to produce green colour to the flame. Take a little of the organic substance on the burnt cu-foil non-luminous flame	The flame is again colored green. If the compound burns with sooty flame.	Halogens present Aromatic compound is present.
4. Take a 5ml of Na-extract solu. In a test tube. Acidify the solution with dil. HNO ₃ Boil for about 5 min. (Boiling is essential to drive out HCN. So if 'N' & 'S' are absent, boiling May be eliminated). After this add a few ml. of AgNO ₃ soln. to mixture.	a) curdy white ppt. Is obtained which is soluble in dil. NH ₄ OH soln. b) A pale yellow ppt. is obtained which dissolves in dil. NH ₄ OH with difficulty but dissolves easily in conc. NH ₄ OH c) Yellow ppt. Obtained and is insoluble in NH ₄ OH soln.	a) Chlorine is present. b) Bromine present. c) Iodine present.
5 a) Take about 5ml of soln. In a test tube. Acidify with acetic acid. Treat the soln. With dil. CCL ₄ and NaNO ₃ shake the mixture thoroughly. b) Acidify the Na extract soln. With dil. HNO ₃ boil for some time, cool and then add a few ml of CCL ₄ to it. After this add an excess of Cl ₂ water and shake.	a) CCL ₄ layer becomes violet. b) CCL ₄ layer becomes brown. 1) CCL ₄ layer becomes violet. 2) CCL ₄ layer becomes violet.	Iodine present. Bromine present. Iodine present. Bromine present.

RESULT:

In the given unknown sample, and are present as special elements.